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Biology Mock

Paper 2 (Elective Part)

60 minutes（1 hour）

FOR REVISION

For revision and self-assessment only. Not an official examination paper.

Difficulty rating：**3.9** / 5.0

INSTRUCTIONS

1. This paper consists of FOUR sections: Section A Human Physiology: Regulation and Control, Section B Applied Ecology, Section C Microorganisms and Humans, Section D Biotechnology.
2. Attempt ALL questions in any TWO sections.
3. Each section carries 20 marks.
4. Illustrate your answers with diagrams wherever appropriate.
5. The diagrams in this paper are not necessarily drawn to scale.

Independent, unofficial mock paper. Not affiliated with any official examination authority.

Section A Human Physiology: Regulation and Control

1.

(a) Table 1 shows the levels of three hormones in the blood of a woman from the age of 44 to 56. She reached menopause at about the age of 51.

表一：一名女士在不同年齡的血液激素水平
Table 1: Blood hormone levels of a woman at different ages

年齡 (歲) Age (years)	促卵泡激素 (IU L^{-1}) FSH (IU L^{-1})	黃體生成素 (IU L^{-1}) LH (IU L^{-1})	雌激素 (pg mL^{-1}) Oestrogen (pg mL^{-1})
44	7	6	140
48	18	12	95
52	52	30	40
56	85	42	18

(i) Describe the changes in the levels of FSH and oestrogen of this woman from the age of 44 to 56. (2 marks) (2)

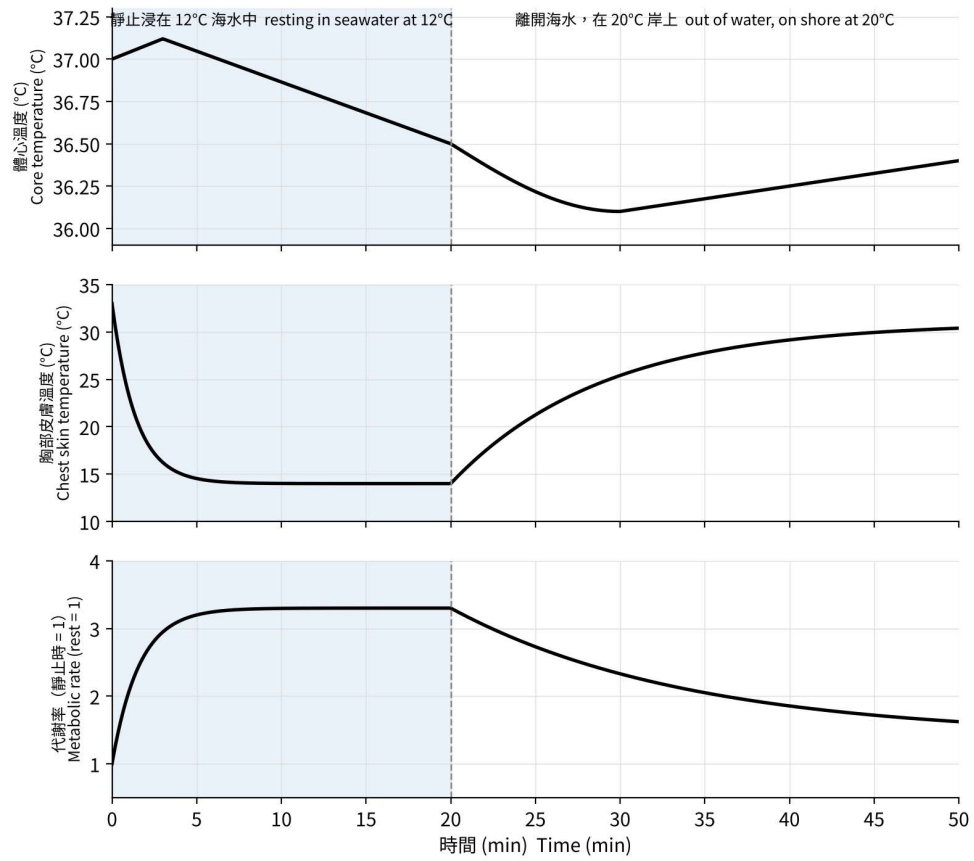
(ii) Explain why the FSH level rises greatly as the oestrogen level falls. (4 marks) (4)

(iii) Hormone replacement therapy (HRT) gives oestrogen (and progesterone) to women at menopause. A doctor said: "All women at menopause should receive HRT." With reference to Table 2, evaluate this statement. (4 marks) (4)

表二：50–59 歲女士接受與沒有接受激素補充療法的比較
Table 2: Comparison of women aged 50–59 with and without HRT

項目 Item	沒有接受激素補充療法 Without HRT	接受激素補充療法 With HRT
每 1000 人在 5 年內髖骨或脊骨骨折的人數 Hip or spine fractures per 1000 women in 5 years	16	10
每 1000 人在 5 年內患乳癌的人數 Breast cancer per 1000 women in 5 years	12	15
每 1000 人在 5 年內出現靜脈血栓的人數 Blood clots in veins per 1000 women in 5 years	5	9
潮熱症狀明顯改善的女士百分率 (%) Women with marked relief of hot flushes (%)	25	80

(b) A winter swimmer rested in seawater at 12°C (head above water, body not moving) for 20 minutes and then returned to the shore at 20°C. The graphs below show his core body temperature, chest skin temperature and metabolic rate.



(i) In the first 5 minutes, chest skin temperature fell from 33°C to about 14°C, but core body temperature changed by only about 0.1°C. Explain how the response of the skin blood vessels maintains the core body temperature. (4 marks) (4)

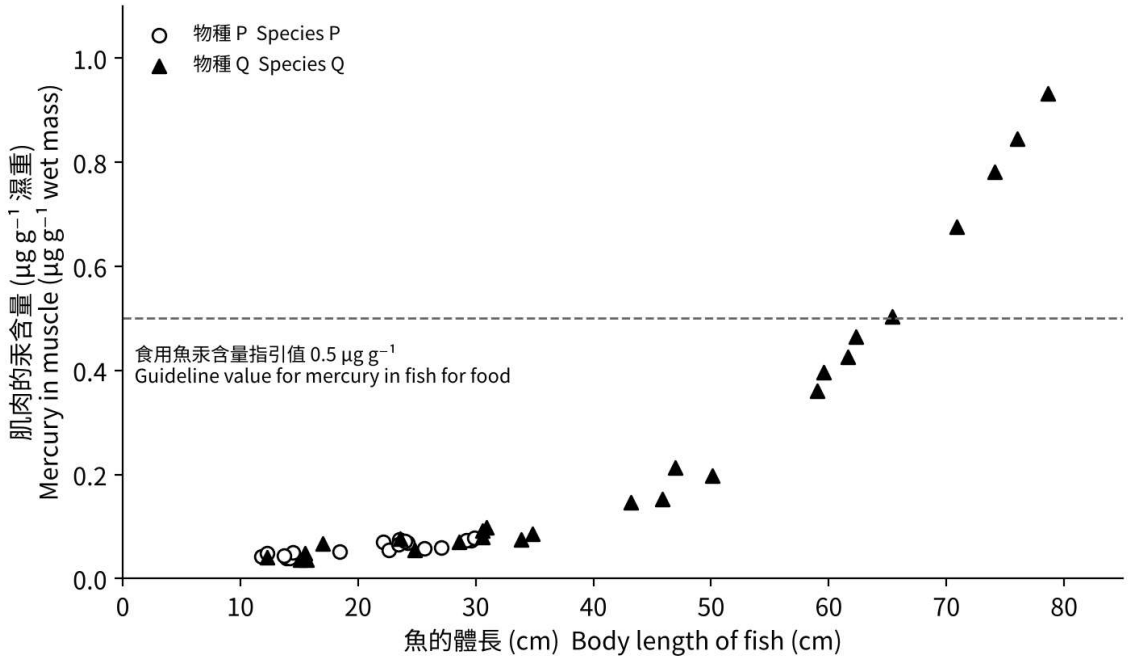
(ii) Describe and explain the change in metabolic rate of this person in the first 5 minutes in the seawater. (3 marks) (3)

(iii) In the first 10 minutes after leaving the water, the core body temperature of this person continued to fall. Suggest an explanation. (3 marks) (3)

Section B Applied Ecology

2.

(a) Some toxic substances (e.g. mercury) are not easily broken down or excreted. They build up in organisms and increase at each level along a food chain (accumulation of toxic substances in food chains). Two species of fish, P and Q, were caught from the same water body. Their body lengths and the mercury content of their muscles were measured. The results are shown below. Species P feeds by filtering plankton throughout its life.



(i) Describe the relationship between mercury content and body length for the two species. (2 marks) (2)

(ii) The researchers think that the diet of species Q changes as it grows. With reference to the data, deduce this change and explain why large individuals of species Q have the highest mercury content. (4 marks) (4)

(iii) A health authority advises pregnant women to "eat 2–3 servings of fish per week, but choose smaller fish that do not prey on other fish". With reference to the data, evaluate this advice. (4 marks) (4)

(b) Researchers built an oyster reef with oyster shells in a bay, allowing wild oysters to settle and grow. The table below shows data from the reef site and a nearby control site without a reef in 2020 (when the reef was built) and in 2025.

蠔礁於 2020 年建於蠔礁地點；對照地點沒有蠔礁
Oyster reef built at the reef site in 2020; no reef at the control site

	對照地點：2020 Control site: 2020	對照地點：2025 Control site: 2025	蠔礁地點：2020 Reef site: 2020	蠔礁地點：2025 Reef site: 2025
懸浮固體 (mg dm^{-3}) Suspended solids (mg dm^{-3})	30	29	31	14
葉綠素 a (μg dm^{-3}) Chlorophyll a (μg dm^{-3})	21	20	22	9
海床水層溶氧量 (mg dm^{-3}) Dissolved oxygen near seabed (mg dm^{-3})	3.9	4.0	3.8	5.9
底棲生物物種數目 Number of benthic species	12	13	11	26

(i) Why did the researchers also collect data from the control site? (2 marks) (2)

(ii) With reference to the data in the table, explain how the oyster reef changed the water quality and the number of benthic species in the bay. (4 marks) (4)

(iii) Someone claimed: "Building oyster reefs can replace sewage treatment works." Evaluate this claim. (4 marks) (4)

Section C Microorganisms and Humans

3.

(a) Sprouts have caused several outbreaks of *Salmonella* food poisoning. Seeds carrying 10^1.0 *Salmonella* per g were germinated at different temperatures for 3 days, and the number of *Salmonella* on the sprouts was measured. The results are shown below.

	種子（萌發前） Seeds (before germination)	在 15°C 萌發 3 日 3 days at 15°C	在 25°C 萌發 3 日 3 days at 25°C	在 35°C 萌發 3 日 3 days at 35°C
沙門氏菌數目 Number of Salmonella ($\log_{10}$ 每克菌落形成單位) ($\log_{10}$ CFU per g)	1.0	3.4	5.9	7.3

(i) Calculate the number of times the *Salmonella* population increased after 3 days of germination at 25°C. (2 marks) (2)

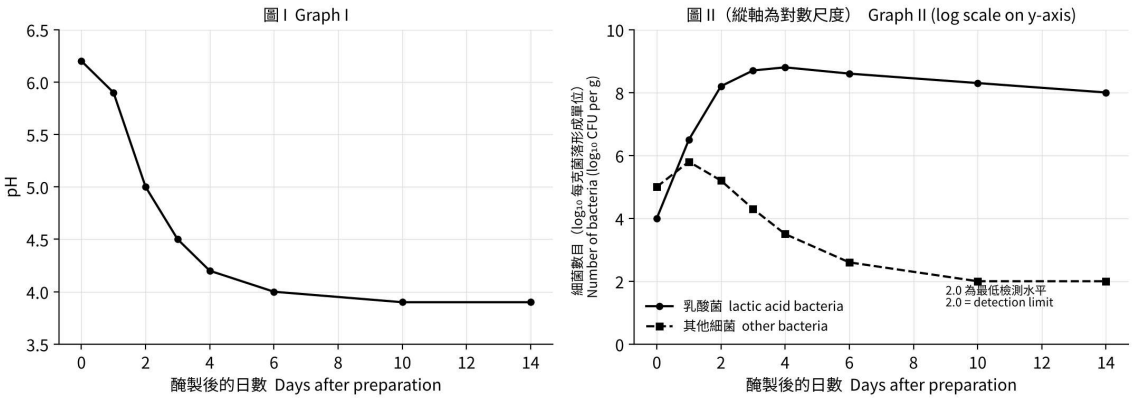
(ii) Explain why sprouts are a high-risk food for food poisoning. (4 marks) (4)

(iii) Two methods of treating seeds were compared in two experiments. The results are shown below. With reference to the data, evaluate which method is more suitable for suppliers, and suggest one measure that consumers can take. (4 marks) (4)

實驗二所用的一般種子處理前帶有 10^1.0 每克沙門氏菌
Ordinary seeds in Expt 2 carried 10^1.0 Salmonella per g before treatment

種子處理方法 Seed treatment	實驗一：人工接種至 10^5.0 每克的種子，處理後的沙門氏菌數目 (log_{10} CFU g^{-1}) Expt 1: seeds inoculated to 10^5.0 per g, Salmonella after treatment (log_{10} CFU g^{-1})	實驗二：萌發率 (%) Expt 2: germination (%)	實驗二：一般種子處理後在 25°C 萌發 3 日，芽菜上的沙門氏菌數目 (log_{10} CFU g^{-1}) Expt 2: ordinary seeds after treatment and 3 days at 25°C, Salmonella on sprouts (log_{10} CFU g^{-1})
沒有處理 untreated	5.0	96	5.9
70°C 熱水浸 90 秒 hot water, 70°C, 90 s	2.0	84	2.6
2% 含氯消毒液浸 15 分鐘 2% chlorine, 15 min	3.2	93	3.9

(b) Pickled vegetables are made by adding salt to vegetables and sealing them in a container for lactic acid fermentation. The graphs below show the changes in pH and the number of bacteria during pickling.



(i) Describe the changes in pH and the number of lactic acid bacteria in the first 4 days. (2 marks) (2)

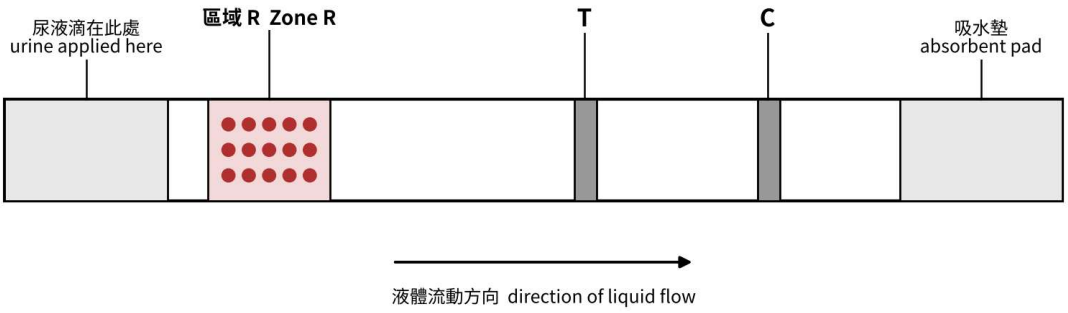
(ii) Explain why the number of other bacteria falls after day 1. (4 marks) (4)

(iii) Someone suggested: (1) adding a little juice from a previous batch to a new batch; (2) halving the amount of salt to make the pickles healthier. With reference to the graphs, evaluate the effects of these two suggestions on the safety of the pickles. (4 marks) (4)

Section D Biotechnology

4.

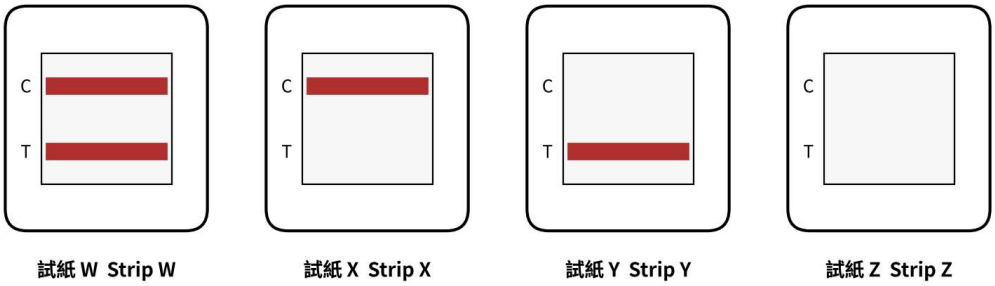
(a) Home pregnancy test strips use monoclonal antibodies to detect human chorionic gonadotrophin (HCG) in urine. The diagram below shows the structure of a strip. Zone R contains anti-HCG antibodies attached to red particles, which can move with the liquid. Lines T and C each carry a type of fixed antibody.



(i) Describe how monoclonal antibodies that bind to HCG are produced. (3 marks) (3)

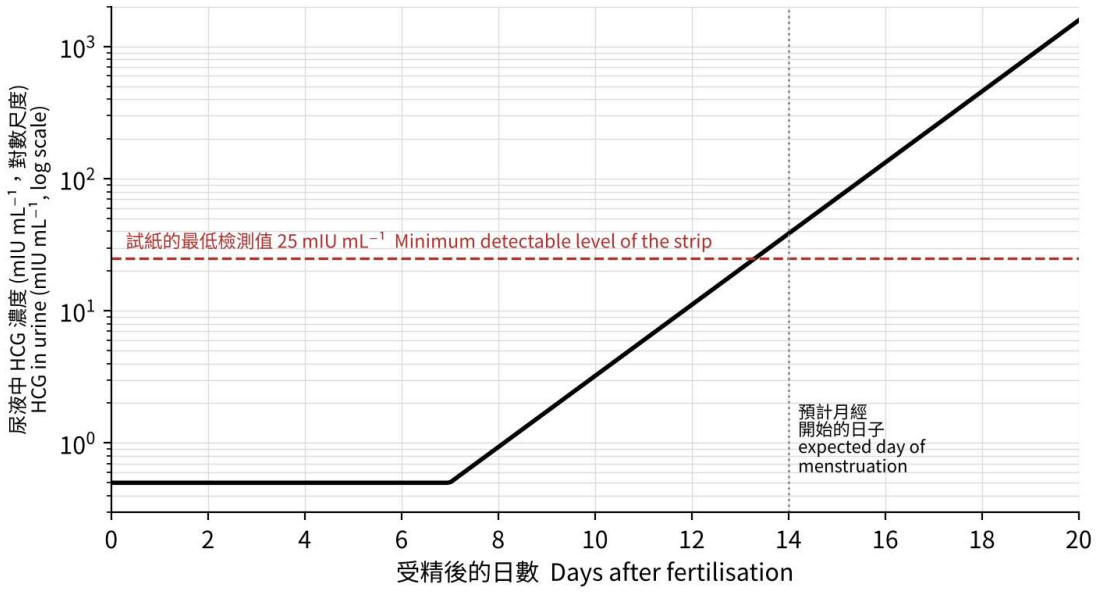
(ii) Line T turns red only when the urine contains HCG. Based on the information given, deduce the property of the antibody fixed at line T and explain how line T turns red. (3 marks) (3)

(iii) The diagram below shows the results of four strips. Interpret the result of each strip. (2 marks) (2)



試紙 W Strip W　試紙 X Strip X　試紙 Y Strip Y　試紙 Z Strip Z

(iv) The graph below shows the change in HCG concentration in urine after fertilisation, together with the minimum detectable level of the strip. With reference to the data, explain why testing too early may give a false negative result, and suggest the earliest time at which testing becomes reliable. (2 marks) (2)



(b) Human growth hormone can be used to treat children who lack it. Before 1985, it was mainly extracted from the pituitary glands of deceased donors; after that, recombinant human growth hormone was produced by genetically modified bacteria. The table below compares the two sources.

	從已死捐贈者的垂體提取 Extracted from pituitary glands of deceased donors	由基因改造細菌生產 Produced by genetically modified bacteria
所得的人長激素 Human growth hormone obtained	每個垂體約 4 mg about 4 mg per gland	每 10 dm^3 培養液約 1 g about 1 g per 10 dm^3 of culture
產品中人長激素的純度 Purity of human growth hormone in product	約 70% about 70%	超過 99% over 99%
已知的問題 Known problems	部分接受者多年後患上致命的腦部疾病 some recipients developed a fatal brain disease years later	未有發現同類問題 no similar problems found

(i) A child needs 18 mg of human growth hormone per month. Calculate how many pituitary glands are needed to treat this child for one year by the extraction method, and use this to state one limitation of the old method. (3 marks) (3)

(ii) With reference to the table, explain the advantages of producing human growth hormone using genetically modified bacteria. (3 marks) (3)

(iii) It has been proposed that human growth hormone be given to healthy children who have normal levels of the hormone but are short. Studies show that with daily injections for five years, these children grow on average only about 4 cm taller as adults. Evaluate this proposal. (4 marks) (4)

END OF PAPER